MATHEMATICAL ANALYSIS 1
HOMEWORK 4

(1) Let $x_0 \in \mathbb{R}$. Recall that $I_r(x_0) = (x_0 - r, x_0 + r)$ is the neighborhood of x_0 of radius $r > 0$, that $I_r^*(x_0) = I_r(x_0) \setminus \{x_0\}$ is the deleted neighborhood of x_0 of radius r, and that $I_M(+\infty) = (M, +\infty)$ is the neighborhood of $+\infty$ of threshold $M > 0$.
   (a) Let $r_1, r_2 > 0$. Prove that $I_{r_1}(x_0) \cap I_{r_2}(x_0)$ is a neighborhood of x_0, and find its radius.
   (b) Let $r_1, r_2 > 0$. Prove that $I^*_{r_1}(x_0) \cap I^*_{r_2}(x_0)$ is a deleted neighborhood of x_0, and find its radius.
   (c) Let $M_1, M_2 > 0$. Prove that $I_{M_1}(+\infty) \cap I_{M_2}(+\infty)$ is a neighborhood of $+\infty$, and find its threshold.

(2) Consider the sequence $\{a_n\}_{n\in\mathbb{N}}$ defined by $a_n = \frac{n^2+(-1)^n n}{n^2+1}$.
   (a) Determine whether the sequence has a finite limit, has an infinite limit, or is indeterminate.
   (b) Prove your answer using the definition of the limit.

(3) Using the definition of the limit, prove that

$$\lim_{n\to\infty} \frac{n^2}{1-2n} = -\infty.$$

(4) Let $\{a_n\}_{n\in\mathbb{N}}$ be a real-valued sequence. Suppose that the limit $\lim_{n\to\infty} |a_n|$ exists. Does $\lim_{n\to\infty} a_n$ exist? If so, prove it. If not, give a counterexample.

(5) The proof of the Monotone Convergence Theorem (Theorem 5.1 in the lecture notes) is written only for the case of an *increasing* sequence. Prove the case of a *decreasing* sequence: let $\{a_n\}_{n\in\mathbb{N}}$ be a real-valued sequence and suppose that there exists $N \in \mathbb{N}$ such that $a_{n+1} \le a_n$ for all $n > N$.
   (a) If $\inf\{a_n \mid n > N\} > -\infty$, then $\lim_{n\to\infty} a_n = \inf\{a_n \mid n > N\}$.
   (b) If $\inf\{a_n \mid n > N\} = -\infty$, then $\lim_{n\to\infty} a_n = -\infty$.

(6) Consider the sequence $a_n = \arctan\left(\frac{5n+6}{n+1}\right)$, $n \in \mathbb{N}$.
   (a) Is this a monotone sequence?
   (b) Find its infimum and supremum (explain your answer using the definitions of infimum and supremum).
   (c) Do the minimum and maximum exist? If so, what are they?
   (d) Does the limit $\lim_{n\to\infty} a_n$ exist? If so, what is it? *Hint: use the previous problem.*

(7) Let $\{a_n\}_{n\in\mathbb{N}}$ be a real-valued sequence. Suppose that it has two subsequences, $\{a_{n_k}\}_{k\in\mathbb{N}}$ and $\{a_{m_k}\}_{k\in\mathbb{N}}$, which converge to two *different* real numbers $\ell_1 \neq \ell_2$. Prove that $\{a_n\}_{n\in\mathbb{N}}$ has no limit (neither a finite limit nor $\pm\infty$). *Hint: argue by contradiction, and treat separately the cases* $a_n \to \ell \in \mathbb{R}$ *(take* $\varepsilon = \frac{|\ell_1-\ell_2|}{4}$*) and* $a_n \to \pm\infty$.

(8) Let $x_n = (-1)^n \frac{2n+1}{n+1}$, $n \in \mathbb{N}$.
   (a) Show that $-2 < x_n < 2$ for all $n \in \mathbb{N}$.
   (b) Using the definition of the limit, show that the subsequence of even terms $\{x_{2k}\}_{k\in\mathbb{N}}$ converges to 2, and that the subsequence of odd terms $\{x_{2k+1}\}_{k\in\mathbb{N}}$ converges to -2.
   (c) Use the previous problem to conclude that the sequence $\{x_n\}_{n\in\mathbb{N}}$ is indeterminate.
   (d) All the terms x_{2k} belong to the open interval $(-2, 2)$, and the sequence $\{x_{2k}\}_{k\in\mathbb{N}}$ converges. Is its limit in $(-2, 2)$? Explain why this does not contradict Proposition 5.3 of the lecture notes, and what Proposition 5.3 says about the limit of $\{x_{2k}\}_{k\in\mathbb{N}}$ when we use the closed interval $[-2, 2]$ instead.

(9) Let $f(x) = \frac{2x^2-3x+1}{x^2+4}$. Find $\lim_{x\to+\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$, and prove your answers using the definition of the limit.

(10) Using the ε-δ definition, prove that $f(x) = 2x^2 + 3$ is continuous at $x_0 = 1$ (that is, $\lim_{x\to 1} f(x) = f(1) = 5$).